

Do **not** write solutions on this page.

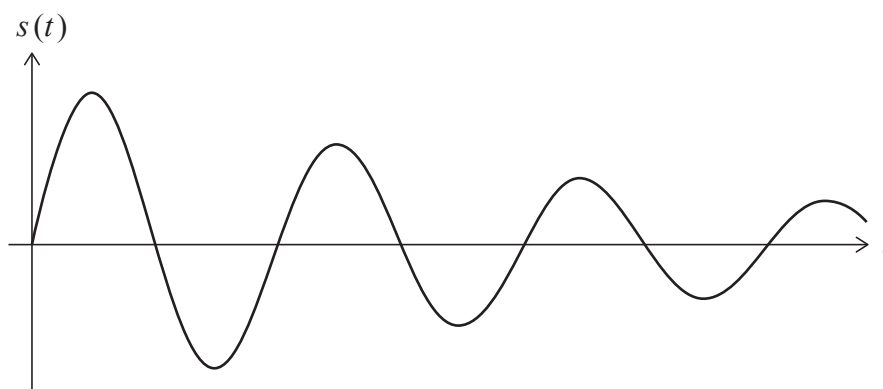
Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

10. [Maximum mark: 17]

A particle P moves in a straight line so that its displacement, s cm, from a fixed point O at time t seconds is given by $s(t) = 2^{\left(1 - \frac{t}{5}\right)} \sin\left(\frac{2\pi t}{3}\right)$, where $t \geq 0$.

The following diagram shows part of the graph of $y = s(t)$.



- (a) Find
- (i) the maximum displacement of P from O ;
 - (ii) the maximum velocity of P . [5]
- (b) Find
- (i) the minimum value of the displacement function $s(t)$;
 - (ii) the displacement of P from O when $t = 3.5$. [3]
- (c) Hence, determine the **total distance** travelled by P in the first 3.5 seconds. [3]
- The first time that P returns and passes through O is when $t = T$.
- (d) Write down the value of T . [1]

(This question continues on the following page)



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(Question 10 continued)

The particle passes through O every T seconds.

A sequence $u_1, u_2, u_3 \dots$ is formed where $u_1, u_2, u_3 \dots$ are the largest **distances** from O in each of the intervals $0 < t < T$, $T < t < 2T$, $2T < t < 3T \dots$ respectively.

It is known that $u_1, u_2, u_3 \dots$ form a geometric sequence.

- (e) (i) Determine the value of the common ratio r of this geometric sequence.
- (ii) Calculate the **total distance** travelled by the particle if it were to continue to move in this way indefinitely.

[5]

